

PCB ASSEMBLY VERIFICATION - AOI REPORT

Inspection Date:	2026-07-25 15:31:06	Operator ID:	Operator_AOI_04
Board Template:	Arduino Uno	Processing Duration:	0.000 sec
Overall Status:	FAIL	Total Components:	14

Missing Components (1)

Component ID	Type	Expected Center (Pct)	Reason
main_mcu_atmega328p	IC	(0.55, 0.45)	Absent from coordinate grid

Misaligned Components (1)

Component ID	Type	Expected Location (Pct)	Actual Location (Pct)	Shift (mm)	Limit (mm)
usb_interface_mcu	IC	(0.22, 0.25)	(0.25, 0.29)	3.25 mm	1.5 mm

Solder / Board Cracks (1)

Crack ID	Target Component	Coordinates (Pct)	Severity
CRK_pin13_led	pin13_led	(0.83, 0.23)	High (Solder Joint Fracture)

Extra Components (1)

Component ID	Type	Coordinates (Pct)	Confidence
ERR_EXTRA_01	Capacitor	(0.15, 0.15)	71.0%